

Functional Verification

Icarus Verilog Regression + Vivado 2024.2 Implementation Sign-off

Regression Test Results

./scripts/run_regression.sh --waves

TEST	핵심 검증 항목	RESULT
tb_common_pkg	spec v2.1 width / depth / count 상수	PASS
..._trace_buffer_basic	reset, ARM, valid gating, early trigger ignore	PASS
..._trace_buffer_trigger	trigger sample alignment, post 512, logical order	PASS
..._trace_buffer_wrap	trigger physical 0 / 1 / 1022 / 1023, wrap order	PASS
..._trace_buffer_control	abort / clear / arm 우선순위, DONE-IRQ hold	PASS
..._trace_buffer_axi	offset, WIP, WSTRB, B/R stall, partial reset	PASS
..._trace_buffer_bram	packaged top, byte 변환, 1,024 Port-B read	PASS
C_Static_assert	register header offset / bit / geometry 일치	PASS

Front-end AXI TB는 Vivado XSim으로 별도 실행 · 원본 VCD 6개는 artifacts/waves/에 저장

RTL Simulation Evidence

VCD · 1 ps resolution



Mandatory ARM Order

Frozen v2.1 §4.1

- | | |
|---|---|
| 1 Sampler Disable 및 divider / mask 설정 | 5 Sampler ENABLE = 1 |
| 2 Trigger Engine CLEAR [아직 ARM 아님] | 6 TRACE_STATUS.PRE_READY = 1 poll |
| 3 Trace Buffer ABORT / CLEAR_DONE | 7 Trigger Engine CLEAR 후 ARM |
| 4 Trace Buffer ARM | 8 TRACE_STATUS.DONE = 1 대기 → BRAM read |

PO 감사 수정 Trigger Engine은 one-shot이므로 **PRE_READY** 확인 전 **ARM**하면 PREFILL 구간에서 pulse가 소진된다.

Implementation Sign-off

Design B · full SoC

Block Design validation	PASS
100 MHz Setup WNS / Hold WHS	+0.832 ns / +0.028 ns
TNS / THS	0.000 ns / 0.000 ns
Failing endpoints DESIGN-WIDE	0 / 9,845
Bitstream DRC error	0
RAMB36E1 32 LMB + 1 CAPTURE	33 / 50
Trace IP routed nets OOC	140 / 140 · error 0

마지막 행만 Trace IP standalone OOC · capture RAMB36E1 = 1은 Tcl assert로 검사